

The τ_σ Involution: A Canonical Galois-Coset Duality on the Binary Icosahedral Group

Lee Smart

Institute of Vibrational Field Dynamics

`contact@vibrationalfelddynamics.org`

May 2026

Status: Preprint (not peer-reviewed). Pure mathematics; no physical interpretation is claimed.

Abstract

We construct an explicit involution $\tau_\sigma : V_{600} \rightarrow V_{600}$ on the vertex set of the 600-cell, equivalently the binary icosahedral group $2I$. The construction is a vertex permutation rather than a Euclidean reflection; testing two restricted families fixing the binary dihedral subgroup $H = \text{Dic}_5$ pointwise — single E_8 -Weyl-reflection candidates (which act trivially on V_{600} in every tested case) and parity-flip candidates (which produce zero closed-and-involution target hits) — motivates using a vertex permutation instead (Remark 3.1; this is a computational observation, not an exhaustive non-existence claim, and $W(H_4)$ was not separately enumerated). The construction instead uses the right-coset T_τ -cycle structure paired with a σ -Galois projection in the icosian trace metric. For each non-bulk right coset $Hg \subset 2I$, which contains exactly one whole T_τ -cycle of K-class 52 and one whole T_τ -cycle of K-class 20 (Paper 1 [Smart, 2026]), the construction selects a T_τ -equivariant phase swap by the symmetric-maximin criterion on the trace inner product $\text{Tr}_{K/\mathbb{Q}} \text{Re}(\sigma(v) \cdot \bar{w})$. The phase selection yields exactly two admissible phases per coset (the antipodal pair $j \in \{0, 5\}$), giving a $\mathbb{Z}_2^5 = 32$ -fold canonical lift. We prove τ_σ satisfies four canonical properties: it is an involution, fixes H pointwise, exchanges the $K = 52$ and $K = 20$ cycle classes within each non-bulk right coset, and commutes with T_τ (T-equivariance). It additionally commutes with the antipodal map $v \mapsto -v$. The canonical phase $(0, 0, 0, 0, 0)$ -lift is exhibited explicitly as a 120-vertex map and verified by an exact-arithmetic certificate.

1 Introduction

The binary icosahedral group $2I$ — equivalently the vertex set V_{600} of the regular 600-cell — admits a rich finite-group structure that has been studied for over a century in the contexts of regular polytopes [Coxeter, 1973], lattice theory [Conway and Sloane, 1988, Elkies, 1995], and Coxeter / reflection groups [Humphreys, 1990]. Recent work [Smart, 2026] has identified a distinguished subgroup chain $H = \text{Dic}_5 \subset 2I$ ($|H| = 20$) arising from the bulk decomposition of V_{600} under T_τ -cycle K-class structure, with H self-normalizing in $2I$.

In this note we construct an involution $\tau_\sigma : V_{600} \rightarrow V_{600}$ that

1. fixes H pointwise,
2. swaps the $K = 52$ and $K = 20$ cycle classes within each non-bulk right coset of H ,
3. commutes with T_τ , and
4. commutes with the antipodal map.

The motivation is internal to the V_{600} programme: a subsequent paper uses this involution to state phase-independence for operator-trace identities on the T_τ -cycle space of V_{600} . The construction is purely combinatorial-arithmetic; no physical interpretation is claimed in the present paper.

A natural first attempt — a Euclidean reflection of V_{600} fixing H — was tested in two restricted families: `block3c_e8_reflection_obstruction.py` reports that single-Euclidean-reflection candidates fixing H act trivially on V_{600} in the tested embedding, and `block3b_parity_flip_tau_sig` reports zero target hits in the parity-flip family. We summarise both in Section 3 (no exhaustive non-existence claim is made, and $W(H_4)$ was not separately enumerated), then turn to a vertex-permutation construction in Section 4. Section 5 proves the four canonical properties of the resulting map; Section 6 discusses the $\mathbb{Z}_2^5$ phase ambiguity.

2 Setup

We import the following from Paper 1 [Smart, 2026]; the present paper does not re-prove any of these statements.

- $K := \mathbb{Q}(\sqrt{5})$ with Galois twist $\sigma : a + b\sqrt{5} \mapsto a - b\sqrt{5}$, extended componentwise to the icosian quaternion algebra $\mathbb{H}_K$.
- $V_{600} = 2I \subset \text{Sp}(1, K)$: the 120 unit icosian quaternions forming the binary icosahedral group, partitioned into 8 axis, 16 half-type, and 96 even-permutation vertices.
- $V_{24} := V_{600} \cap \text{Fix}(\sigma)$: the 24-cell sublattice (8 axis + 16 half-type vertices).
- $T_\tau : g \mapsto \tau g$ for some chosen $\tau \in 2I$ with $\text{ord}(\tau) = 10$. The cycle decomposition of T_τ partitions V_{600} into 12 cycles each of length 10, with K-multiset $\{72:1, 0:1, 52:5, 20:5\}$.
- $H := \text{Dic}_5$, the bulk subgroup of order 20 formed by the union of the $K = 72$ and $K = 0$ cycles. Self-normalizing and non-normal in $2I$ [Smart, 2026].

Lemma 2.1 (Right-coset carrier; Paper 1, Coset cycle composition). *For every non-bulk right coset $Hg \subset 2I$, the coset is exactly the disjoint union of one whole T_τ -cycle of K -class 52 and one whole T_τ -cycle of K -class 20. In particular T_τ preserves Hg setwise (since $\tau \in H$, where τ is the chosen order-10 generator of the bulk T_τ -cycle of $K = 72$ that lies inside $H = \text{Dic}_5$, we have $\tau H = H$ and hence $T_\tau(Hg) = (\tau H)g = Hg$); the K -class pairing 52 with 20 is the additional content of the lemma and is established in Paper 1 by direct enumeration. The five non-bulk right cosets exhaust the ten non-bulk T_τ -cycles, paired one $K = 52$ with one $K = 20$.*

By contrast, non-bulk *left* cosets gH contain two vertices from each of the ten non-bulk T_τ -cycles and are *not* whole-cycle carriers (Paper 1, “Coset cycle composition” lemma). For the construction below it is essential to use the right-coset family.

2.1 Trace inner product

The icosian quaternion algebra $\mathbb{H}_K$ with $K = \mathbb{Q}(\sqrt{5})$ carries a natural rational trace bilinear form,

$$\langle v, w \rangle_{\text{tr}} := \text{Tr}_{K/\mathbb{Q}}(\text{Re}(v \cdot \bar{w})),$$

where $\bar{w}$ is the quaternionic conjugate, Re extracts the real (scalar) component, and $\text{Tr}_{K/\mathbb{Q}}(a + b\sqrt{5}) = 2a$. Up to a global rescaling, this form puts V_{600} inside the E_8 lattice [Elkies, 1995, Conway and Sloane, 1988]; on the cross-Galois pairs used in the construction below the values are rational (and in fact lie in $\frac{1}{4}\mathbb{Z}$), and all arithmetic is exact in $\mathbb{Q}(\sqrt{5})$.

3 Why Euclidean reflections fail

The natural first attempt at constructing τ_σ is a Euclidean reflection of V_{600} fixing H pointwise. The following computational observation motivates abandoning this route.

Remark 3.1 (Reflection obstruction, computational). *Two auxiliary script families in the V_{600} programme repository test isometric τ_σ candidates that fix $H = \text{Dic}_5$ pointwise:*

- `block3c_e8_reflection_obstruction.py` (with the embedding helpers in `block3a_e8_icosian_lattice.py`) checks single Euclidean-reflection candidates fixing H and reports that each such candidate acts trivially on V_{600} in the tested embedding;
- `block3b_parity_flip_tau_sigma.py` searches the parity-flip family for a V_{600} -permutation that fixes H pointwise, swaps the $K = 52$ and $K = 20$ cycle classes, and is closed under $2I$, and reports zero target hits.

No exhaustive Weyl-group enumeration is claimed, and we do not assert a closed-form non-existence theorem; this remark only motivates the vertex-permutation construction below.

The remainder of the paper does not depend on Remark 3.1 mathematically; it is included only to motivate why we construct τ_σ as a vertex permutation rather than as a reflection. The construction proceeds via a σ -Galois projection on right cosets.

4 The σ -projection construction

4.1 Cycle phase swap

For a non-bulk right coset $Hg \subset 2I$, fix base points $v_0 \in C_{52}$ and $w_0 \in C_{20}$ (both supplied by the deterministic vertex enumeration of `vfd_v600.icosian.build_vertices()`; see Section 6 for the convention dependence) and write the constituent T_τ -cycles as

$$C_{52} = (v_0, v_1, \dots, v_9), \quad C_{20} = (w_0, w_1, \dots, w_9),$$

with $v_{k+1} = \tau v_k$ and $w_{k+1} = \tau w_k$ (indices mod 10). The phase labels below are computed relative to this choice; a different pair (v_0, w_0) would relabel them.

Definition 4.1 (Phase- j swap). *For each $j \in \mathbb{Z}/10$, the phase- j swap is the involution $\tau_\sigma^{(j)} : Hg \rightarrow Hg$ defined by*

$$\tau_\sigma^{(j)}(v_k) := w_{(j+k) \bmod 10}, \quad \tau_\sigma^{(j)}(w_{(j+k) \bmod 10}) := v_k.$$

Each $\tau_\sigma^{(j)}$ exchanges the two cycles, is a well-defined involution on Hg , and is T_τ -equivariant by construction.

There are 10 phases per coset, so naively $10^5 = 10^5$ candidates across the five non-bulk cosets. The σ -projection criterion below restricts to $2^5 = 32$ canonical lifts.

4.2 Symmetric-maximin σ -projection

Definition 4.2 (Symmetric maximin score). *For a non-bulk right coset Hg with cycle data $(C_{52}, C_{20}) = (v_k, w_k)$, define the forward and backward σ -projections*

$$s_{\rightarrow}(j, k) := \langle \sigma(v_k), w_{(j+k) \bmod 10} \rangle_{\text{tr}}, \quad s_{\leftarrow}(j, k) := \langle \sigma(w_{(j+k) \bmod 10}), v_k \rangle_{\text{tr}},$$

and the symmetric phase score

$$s(j) := \min_{k \in \mathbb{Z}/10} \min(s_{\rightarrow}(j, k), s_{\leftarrow}(j, k)).$$

The maximin admissible phase set is

$$\Phi(Hg) := \operatorname{argmax}_{j \in \mathbb{Z}/10} s(j).$$

The two-sided minimum makes $s(j)$ symmetric in the role of the two cycles and is the form verified by the companion script `verify.py`.

Theorem 4.1 (Phase selection). *Under the base-point convention of Section 4 (both $v_0 \in C_{52}$ and $w_0 \in C_{20}$ supplied by the deterministic ordering of `vfd_v600.icosian.build_vertices()`), for every non-bulk right coset $Hg \subset 2I$, $\Phi(Hg) = \{0, 5\}$, with $s(0) = s(5) = 0$ and $s(j) = -\frac{5}{4}$ for $j \in \{1, 2, 3, 4, 6, 7, 8, 9\}$. The two winning phases are separated by 5 (i.e., antipodal indices on the length-10 cycle); the labels $\{0, 5\}$ themselves are convention-dependent (Remark 6.1). Phase $j = 5$ is the antipodal shift of $j = 0$.*

Proof. Direct exact-arithmetic computation in $\mathbb{Q}(\sqrt{5})$ (the companion script `verify.py` enumerates all five non-bulk right cosets and all 10 phases, asserts the full symmetric score table $s(0) = s(5) = 0$ and $s(j) = -\frac{5}{4}$ otherwise, and hence verifies the `argmax`). The identity $j = 5 \leftrightarrow j = 0 + \text{antipode}$ follows from the central involution $\tau^5 = -1 \in 2I$ acting on cycle indices by $k \mapsto k + 5 \bmod 10$. $\square$

4.3 Canonical lift

Definition 4.1 and Theorem 4.1 together yield $|\Phi(Hg)|^5 = 2^5 = 32$ admissible compositions across the five non-bulk right cosets. We name the all-zero choice canonical.

4.4 Algorithm (pseudocode)

The construction and verification can be summarised as the following explicit pseudocode (the script `verify.py` is a literal translation):

```

build state via vfd_v600.group.build_state()
  -> verts, cycles, K_of_cycle, Dic5, T_tau

for each non-bulk right coset Hg of H = Dic_5:
  let C52 = the K=52 cycle in Hg, base point v_0
  let C20 = the K=20 cycle in Hg, base point w_0
  for j in 0..9:
    s(j) := min over k in 0..9 of
      min( <sigma(v_k), w_{(j+k) mod 10}>_tr,
           <sigma(w_{(j+k) mod 10}), v_k>_tr )
  Phi(Hg) := argmax_{j} s(j)          # equals {0, 5}

choose phase j_g = 0 for each Hg (canonical lift)
build tau_sigma:
  tau_sigma(h)      := h              for h in Dic_5
  tau_sigma(v_k)    := w_{(j_g+k) mod 10}
  tau_sigma(w_{j_g+k}) := v_k

verify: P1 (involution), P2 (fix Dic_5),
        P3 (K=52 <-> K=20), P4 (T_tau-equivariance),
        antipodal: tau_sigma(-v) == -tau_sigma(v),
        right-coset preservation,
        match against data/tau_sigma_canonical.txt.
```

Definition 4.3 (Canonical τ_σ). *The canonical τ_σ is the involution $\tau_\sigma : V_{600} \rightarrow V_{600}$ obtained by:*

- $\tau_\sigma(h) = h$ for every $h \in H$;
- For each of the five non-bulk right cosets Hg , applying the phase-0 swap $\tau_\sigma^{(0)}$ of Definition 4.1.

The 120-vertex map is recorded explicitly in `papers/v600-programme/data/tau_sigma_canonical.txt`.

4.5 Worked example

For the canonical state with τ chosen as the first order-10 icosian in the deterministic vertex enumeration, the first non-bulk right coset has $K = 52$ cycle and $K = 20$ cycle

$$C_{52} = [4, 119, 81, 80, 118, 5, 112, 86, 87, 113], \quad C_{20} = [6, 93, 59, 58, 92, 7, 90, 60, 61, 91].$$

The phase-0 swap pairs them index-wise: $4 \leftrightarrow 6, 119 \leftrightarrow 93, 81 \leftrightarrow 59, 80 \leftrightarrow 58, 118 \leftrightarrow 92, 5 \leftrightarrow 7, 112 \leftrightarrow 90, 86 \leftrightarrow 60, 87 \leftrightarrow 61, 113 \leftrightarrow 91$.

5 Main theorem

Theorem 5.1 (Canonical τ_σ properties). *The canonical τ_σ of Definition 4.3 satisfies:*

- (P1) Involution: $\tau_\sigma \circ \tau_\sigma = \text{id}_{V_{600}}$.
- (P2) Fixes the bulk subgroup: $\tau_\sigma(h) = h$ for every $h \in H$.
- (P3) Cross-K coset swap: for every $v \in Hg$ with Hg non-bulk, the K -class of the cycle of $\tau_\sigma(v)$ is the K -opposite of the K -class of the cycle of v (i.e., $52 \leftrightarrow 20$).
- (P4) T_τ -equivariance: $\tau_\sigma \circ T_\tau = T_\tau \circ \tau_\sigma$.

Additionally, τ_σ commutes with the antipodal map $v \mapsto -v$.

Proof. (P1) Each phase-0 swap on a non-bulk right coset is an involution by Definition 4.1, and identity is an involution on H ; the disjoint union of involutions on the six cosets is an involution on V_{600} .

(P2) Direct from the definition.

(P3) Each phase-0 swap exchanges C_{52} and C_{20} in its right coset; the K -class assignment is T_τ -cycle-determined, so each swap pair has K -classes $\{52, 20\}$.

(P4) For $v = v_k \in C_{52}$ in some non-bulk right coset, $\tau_\sigma(T_\tau(v_k)) = \tau_\sigma(v_{k+1}) = w_{k+1} = T_\tau(w_k) = T_\tau(\tau_\sigma(v_k))$, with indices mod 10. Same for $v = w_k$. For $v \in H$, $\tau_\sigma(v) = v$ and $T_\tau(v) = \tau v \in H$ also fixed.

Antipodal compatibility. The element $\tau \in 2I$ has order 10 in the binary icosahedral group; the unique central involution of $2I$ is -1 , and the order-10 subgroup $\langle \tau \rangle$ contains it as $\tau^5 = -1$ (this is the standard unit-quaternion fact $|\tau| = 1$ with $\tau^{10} = 1$ forcing $\tau^5 \in \{\pm 1\}$; $\tau^5 = 1$ would contradict $\text{ord}(\tau) = 10$). Hence the antipodal map acts on cycle indices by $v_k \mapsto -v_k = \tau^5 v_k = v_{k+5}$ (and similarly $w_k \mapsto w_{k+5}$), with indices mod 10. Both v_{k+5} and w_{k+5} remain in the same right coset Hg , since right cosets are setwise T_τ -invariant by Lemma 2.1. Therefore

$$\tau_\sigma(-v_k) = \tau_\sigma(v_{k+5}) = w_{k+5} = -w_k = -\tau_\sigma(v_k),$$

and analogously for $v = w_k$. For $v \in H$, $\tau_\sigma(v) = v$ and $-v \in H$ (since $-1 \in H$), so $\tau_\sigma(-v) = -v = -\tau_\sigma(v)$. $\square$

6 The $\mathbb{Z}_2^5$ phase ambiguity

By Theorem 4.1, each non-bulk right coset admits two maximin phases, $\{0, 5\}$. Choosing a phase independently for each of the five non-bulk right cosets gives $2^5 = 32$ admissible involutions, all satisfying (P1)–(P4) of Theorem 5.1. The group of phase choices is $\mathbb{Z}_2^5$, generated by the five per-coset antipodal shifts $j \mapsto j + 5$.

Remark 6.1 (Convention vs. freedom). *The labels $\{0, 5\}$ for the maximin phase set are convention-dependent; what is intrinsic is that $|\Phi(Hg)| = 2$ and that the two winning phases are antipodal (separated by 5 on the length-10 cycle, equivalently related by $\tau^5 = -1$). The specific labels arise from: (i) the choice of base points $v_0 \in C_{52}$ and $w_0 \in C_{20}$ in Definition 4.1; and (ii) the deterministic ordering of vertices supplied by `vfd_v600.icosian.build_vertices()`. A different choice of base points relabels the phases by a shift $j \mapsto j + c$ for some $c \in \mathbb{Z}/10$, but the $\mathbb{Z}_2^5$ orbit of canonical lifts is intrinsic to the construction.*

The certificate `verify.py` enumerates all 32 phase combinations and confirms that every one yields an admissible τ_σ involution satisfying (P1)–(P4).

7 Conclusion

The binary icosahedral group $2I$ admits a canonical σ -Galois involution τ_σ satisfying four structural properties (P1)–(P4) of Theorem 5.1, plus antipodal compatibility. The construction is a vertex permutation (motivated, but not required, by the computational reflection obstruction recorded in Remark 3.1), realised via cycle-phase swaps on right cosets of the bulk subgroup Dic_5 selected by symmetric-maximin σ -projection in the icosian trace metric (Theorem 4.1). The $\mathbb{Z}_2^5$ ambiguity in phase choice yields 32 canonical lifts; we name the all-zero choice the canonical representative and exhibit its 120-vertex map explicitly.

A Verification certificate

The companion script `verify.py` performs the following exact checks:

1. $|V_{600}| = 120$, $|H| = 20$, $|V_{24}| = 24$.
2. Every non-bulk right coset Hg is one whole T_τ -cycle of $K = 52$ plus one whole T_τ -cycle of $K = 20$.
3. Regression: every non-bulk left coset gH contains exactly two vertices from each of the ten non-bulk T_τ -cycles (*not* a whole-cycle carrier).
4. Per non-bulk right coset, the symmetric maximin score $s(j)$ of Definition 4.2 satisfies $s(0) = s(5) = 0$ and $s(j) = -\frac{5}{4}$ for $j \in \{1, 2, 3, 4, 6, 7, 8, 9\}$, hence $\Phi(Hg) = \{0, 5\}$.
5. The phase-(0, 0, 0, 0, 0) rebuild matches the loaded canonical map `data/tau_sigma_canonical.txt` vertex-by-vertex.
6. The rebuilt map satisfies (P1)–(P4) and antipodal compatibility.
7. Right-coset preservation: τ_σ preserves each right coset Hg setwise (post-build check, not part of the theorem statement; it holds by Definition 4.3, since each per-coset phase swap is internal to its right coset, and is checked separately by the script — properties (P1)–(P4) alone do not imply it, since one could in principle pair a $K = 52$ cycle of one coset with a $K = 20$ cycle of another and still satisfy (P1)–(P4)).
8. All $2^5 = 32$ phase combinations yield valid involutions satisfying (P1)–(P4).

The script depends only on the Python standard library and the shared `vfd_v600` package. Reproducibility command:

```
PYTHONPATH=papers/v600-programme/lib \  
python3 papers/tau-sigma-construction/verify.py
```

References

John H. Conway and Neil J. A. Sloane. *Sphere Packings, Lattices and Groups*. Springer-Verlag, 3rd edition, 1988.

H. S. M. Coxeter. *Regular Polytopes*. Dover, 3rd edition, 1973.

Noam D. Elkies. E_8 lattice and Galois theory. *Mathematical Intelligencer*, 1995.

James E. Humphreys. *Reflection Groups and Coxeter Groups*. Cambridge University Press, 1990.

Lee Smart. A $1/4$ incidence identity on the 600-cell and its Bekenstein–Hawking analogy. V_{600} programme, Paper 1., 2026.